The Verifier as the Sampler: Certified Frequency Control on the Lossy 39-Bus Grid

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Abstract

Frequency control holds a grid at an operating point, so a stability certificate for a learned controller is only as good as the region it covers around that point. On the lossy Kron-reduced New England 39-bus model we reproduce a sample-based Lyapunov pipeline and measure where it fails, finding that its value function violates the decrease condition on 49.9% of a dense grid spanning the one-sigma operating envelope (maximum Lie derivative $+36$), while 200,000 uniform random draws report a clean bill of health and retraining toward the operating point does not repair it. We then put the verifier inside the training loop in place of the sampler. CROWN branch-and-bound supplies region-wide bounds and localized counterexamples instead of random draws, and positive definiteness is written into the value function structurally as a sum of squares with $V(z^\star) = 0$ exactly. The result is a certificate on the operating annulus for all five seeds, at exponential rate $\beta \geq 4.0$ and zero tolerance, confirmed by a projected-gradient audit, an independent Jacobian path, and the SMT solver dReal. We also measure the boundary of the method, where the 4-D ring fails with a genuine counterexample and the full 20-D state fails at radius 0.01, and report every measured cost. The broader lesson for physical-system learning is that a certificate is a statement about a region, so a verification loop which chooses that region is a different object from a sampler which cannot, and the gap between the two is where the failures of a learned controller actually hide.

1 Introduction

A power grid stays up because its generators stay synchronized, and they stay synchronized because frequency control pulls them back to a shared operating point after every disturbance. As wind and solar displace the synchronous machines whose rotating mass supplies that inertia, the quantity deciding how much renewable generation a network absorbs is the size of the set of post-fault states its controller returns to synchronism, and a stability certificate is exactly that quantity when the region it certifies is the region the grid runs in.

The classical certificate for this problem is an energy function, and it does not exist for a network with resistive losses, because the lossy term $G_{ij}\cos(\delta_i - \delta_j)$ is not the gradient of any scalar potential [Chiang, 1989]. Prior work works around the obstruction by learning a neural Lyapunov function with a decentralized controller and treating sampled violations of the conditions as a training loss [Cui and Zhang, 2022], and that line closes with the gap it leaves open, which is whether the learned function satisfies the conditions for *every* state in a region rather than at the points it was trained on. More samples do not close it, since a certificate is universally quantified over a continuum and any finite sample set has measure zero.

This paper asks what a certificate has to cover and whether it transfers, and three measured claims carry the answer. First, the region a sample-based pipeline certifies is not the region the grid operates in, since its value function violates the decrease condition on 49.9% of a dense grid over the envelope while a 200,000-draw random audit finds nothing, and recentering does not fix it. Second, the repair is architectural rather than statistical, since positive definiteness written into the value function as a sum of squares holds for every parameter value by construction. Third, the wall is measured rather

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than hidden, because the certificate does not transfer from a 2-D ring to a 4-D ring, where a genuine counterexample survives dense evaluation on every seed.

2 Problem formulation

Plant and controller. The New England 39-bus system is Kron-reduced to $n = 10$ generator buses, which eliminates the algebraic load buses and leaves a differential model in rotor angles and frequencies alone. With the state $x = (\delta, \omega) \in \mathbb{R}^{2n}$, where δ_i is the rotor angle deviation of generator i in radians and ω_i its frequency deviation in per-unit Hz, the lossy swing dynamics are

$$\dot{\delta}_i = \omega_{\text{scale}}\, \omega_i, \qquad M_i \dot{\omega}_i = P_i - u_i(\omega_i) - \textstyle\sum_j B_{ij} \sin(\delta_i - \delta_j) - \sum_j G_{ij} \cos(\delta_i - \delta_j), \tag{1}$$

with $\omega_{\text{scale}} = 2\pi$. Here $M_i > 0$ is inertia, P_i the net mechanical input, B_{ij} and G_{ij} the susceptance and conductance coupling between generators i and j, and u_i the primary frequency control action. The conductance terms are the lossy part, and they are what remove the scalar potential. The certified controller is the saturated droop law $u_i(\omega_i) = \text{sat}(c_i \omega_i)$ with droop coefficient c_i, which is piecewise linear, so a verifier can bound it exactly rather than relaxing it. We work in reference-subtracted coordinates $z = [\delta_i - \delta_1,\ \omega_i - \omega_1]$ for $i = 2, \ldots, n$, which removes the rotational invariance of the angles and makes the synchronous equilibrium an isolated point $z^\star$ whose counterpart in the original coordinates is $x^\star$. Regions below are described in z and reported in δ_5 and ω_5.

Lyapunov conditions. Let $V_\phi : \mathcal{D} \to \mathbb{R}$ be a continuously differentiable candidate parameterized by the network weights ϕ, with $V^\star = V_\phi(z^\star)$ its value at the equilibrium. The candidate certifies when two conditions hold: positive definiteness,

$$V_\phi(z) - V^\star > 0 \tag{4a}$$

and decrease,

$$\dot{V}_\phi(z) = \nabla V_\phi(z)^\top f_{\text{cl}}(z) < 0 \tag{4b}$$

for all $z \neq z^\star$ in the region of interest, where f_{cl} is the closed-loop vector field of (1). We also certify the exponential strengthening $\dot{V}_\phi(z) \leq -\beta\,(V_\phi(z) - V^\star)$ for a rate $\beta > 0$, which is the stronger statement and the one an operator can turn into a settling-time bound.

Sound region verification. Verification replaces pointwise evaluation with a bound that holds everywhere on a set. With $G_\phi(z) = V_\phi(z) - V^\star$ the positive-definiteness margin and $F_\phi(z) = -\dot{V}_\phi(z)$ the decrease margin, we build one computation graph containing V_ϕ, its analytic gradient, the controller, and the sine and cosine nonlinearities of (1), and CROWN propagates a linear relaxation of it to a sound upper bound on G_ϕ and F_ϕ over an input box [Zhang et al., 2018, Xu et al., 2020], while branch-and-bound splits inconclusive boxes [Shi et al., 2024a]. A box certifies when the lower bound on G_ϕ is strictly positive and the decrease margin is nonnegative on every sub-box, and a feasible state with a negative margin is retained as a counterexample while a box that exhausts its budget is reported as `unknown` rather than as a refutation.

3 The verifier as the sampler

The method follows from one observation about sampling. A certificate quantifies over a region while a sampler visits a measure-zero set, so the sampler that matters for certification is the verifier, and both training interfaces below use it. In the counterexample-guided interface, every feasible counterexample from a failed subregion is appended to the training buffer and the network is retrained against it with the controller held fixed. In the certified-training interface the CROWN lower bound is differentiable with respect to ϕ, so the bound enters the loss and raises the worst-case margin over the box rather than at sampled points [Shi et al., 2024b]. Both interfaces are training only, and the certificate is the final branch-and-bound pass over the finished network, which Figure 1 draws as the loop it is.

Positive definiteness is where soft constraints fail, and the failure is measurable. The as-trained network violates condition (4a) outright on the target annulus, with a verified verdict of `violation` and a counterexample margin of -0.008, and retraining leaves (4a) in active tension with the exponential condition near the equilibrium because the two losses pull on the same parameters. The fix is to stop training that property and write it into the architecture,

$$V(z) = \|g(z) - g(z^\star)\|^2 + c\,\|z - z^\star\|^2, \qquad c > 0, \tag{2}$$

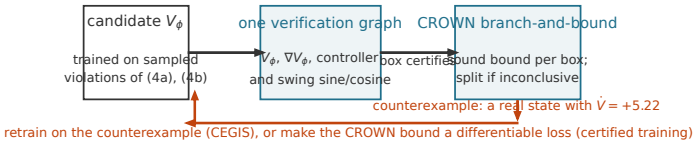


Figure 1: The verifier-as-sampler loop. The closed loop and the candidate V_ϕ live in one computation graph that CROWN branch-and-bound bounds over the target region, splitting any subregion whose bound is inconclusive, and a failed subregion returns a sound, localized counterexample that either refines the training buffer or enters the loss through the differentiable bound. Only the final branch-and-bound pass is a proof.

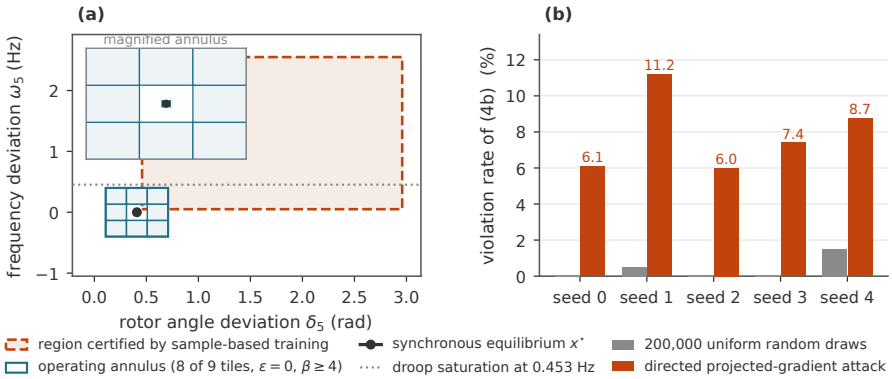


Figure 2: Where the certificate sits, and what sampling misses. Panel (a) is the generator-5 plane with the equilibrium marked. The dashed box is what certified training certifies, offset 0.05 off the equilibrium and running to 2.55 Hz, well past droop saturation at 0.453 Hz, and the shaded 3×3 tiles are the operating annulus this paper certifies instead, with the centre cell removed because strict decrease is structurally impossible at the equilibrium. Panel (b) compares violation rates of the decrease condition under 200,000 uniform random draws against a directed projected-gradient attack, per seed, where the draws find essentially nothing and the directed search finds 6 to 11%.

where g is the learned feature map. Then $V(z^\star) = 0$ exactly and $V(z) \geq c\|z - z^\star\|^2$ for *every* parameter value, so (4a) holds by construction, no training run can break it, and no verifier has to spend budget proving it. That is the difference between a property enforced structurally and one sampled well, and it is the single change that makes the operating-point certificate possible.

4 Experiments

Results are single-CPU, multi-seed, and regenerated from saved per-seed records.

Result 1: sampling does not see the failure, and the certified region is not where the grid runs. On the as-trained network none of 200,000 uniform random states violates (4b), while a directed projected-gradient attack finds violations from 6.1% to 11.2% of its starts with a maximum Lie derivative near$+200$, so the violation set is real and invisible to the sampler. The harder finding is positional, and Figure 2a shows both regions in the generator-5 plane, where the box that certified training certifies sits offset from the equilibrium by 0.05 and reaches 2.55 Hz, past droop saturation at 0.453 Hz, so it covers a region the grid does not operate in and only clips the one it does. Evaluating that checkpoint densely over the one-sigma envelope, 1681 points with the other coordinates pinned, finds (4b) violated at 49.9% of them with a maximum Lie derivative of $+35.98$, and retraining with the box centered on the equilibrium does not repair it, because nothing certifies at scale 1.0 even at tolerance 10^{-3} and the smallest achievable tolerance is 0.0156. The failure is neither the verifier nor the sampler but the soft-constrained value function, which is why (2) is the fix rather than a better sampling scheme.

Result 2: the operating annulus certifies on every seed. Strict decrease is structurally impossible on any region containing the equilibrium because $\dot{V} = 0$ there exactly, so the certificate is a punctured-region statement, and the hole is not an uncertified gap, because stability inside it is the standard local

Table 1: Certification on the operating annulus from five saved per-seed records, with the audited decrease margin $F_\phi(z) = -\dot{V}_\phi(z)$ in units of 10^{-3}. The rate $\beta = 4.0$ is a lower bound set by the tolerance grid, and the last row is the independent check.

Quantity	seed 0	seed 1	seed 2	seed 3	seed 4
Tiles certified, $\varepsilon = 0$	8/8	8/8	8/8	8/8	8/8
Audited margin F_ϕ (10^{-3})	0.94	0.99	1.52	1.02	0.75
Certified rate β	4.0 on every seed, a lower bound				
Jacobian path, dReal	no contradiction in 24 of 24 boxes; `unsat` on 8 of 8 tiles, seeds 0 to 2				

conclusion from the linearization, while the certificate is what extends the guarantee to the annulus around it.The operating box of generator 5 is $\delta_5 \in [\delta_5^\star \pm 0.3]$ rad and $\omega_5 \in [\pm 0.4]$ Hz, cut into a 3×3 grid with the centre cell, a ± 0.015 rad and ± 0.02 Hz neighbourhood of the equilibrium, removed. The eight remaining tiles certify individually and the box stays inside linear droop saturation, so the certificate covers the unsaturated excursions normal operation occupies. Table 1 reports the outcome, and all eight tiles certify at zero tolerance on all five seeds with a certified rate $\beta \geq 4.0$, a lower bound set by the tolerance grid rather than by the true worst case. Every certificate passes an independent projected-gradient audit with minimum margin 7.5×10^{-4}, a soundness scan in which the analytic-gradient and independent Jacobian paths never contradict across 24 boxes, and the δ-complete SMT solver dReal at $\delta = 10^{-3}$, which returns `unsat` on all eight tiles for the three seeds checked. Smaller excluded neighbourhoods returned `unknown` under the budget, so the reported hole is the smallest that certifies rather than the smallest that exists.

The learned controller carries the certificate. Certifying the droop loop is necessary but not sufficient, because a certificate that only holds for a stand-in law does not speak to the deployed policy, so we reproduce the monotone stacked-ReLU controller of the method we build on, with its Lyapunov regularizer, and rerun certification under it. The certificate holds at $\rho = 2.0$, the same region as under droop, so the certified sublevel set is forward invariant under the learned policy and not only under the law used to train V. The cost is reported as measured, and it is negative, because the 19% cost reduction reported by that method is tied to a line trip whose post-fault network is not in the shipped data, so we model the disturbance as an initial perturbation instead, and in that regime the regularized controller costs 35.6% more than droop, at 0.943 against 0.695, so on this measure the learned policy does not yet pay for itself and the certificate result does not depend on it. Under a large modeled kick both controllers lose synchronism with a final deviation near 5.7 Hz, so the certificate extends to the learned policy over its certified region and no further.

5 Discussion and limitations

The scope of the headline result has to be stated precisely, since the certificate in Table 1 is a two-dimensional slice with the remaining eighteen coordinates pinned, so it states a property of that slice rather than of the full power system, and we make no grid-stability claim beyond it. On offset boxes certified training reaches $\rho = 2.5$ on all five seeds in both 2-D and 4-D, always with the equilibrium excluded, and at the operating point the climb stops. The 4-D punctured box, with generators 5 and 9 active, does not certify, and dense evaluation finds a genuine exponential-condition violation on every seed with a minimum margin between -0.050 and -0.057, while the structural positive-definiteness condition stays clean. In the full 20-D state the slice-hardened network genuinely violates (4b) even at radius 0.01. Those are counterexamples rather than verifier timeouts, and publishing them tells the next attempt where to spend effort.

Limitations. The remaining limits are exact. The equilibrium is fixed and the plant is Kron-reduced, so nothing here speaks to switching topology or to the algebraic dynamics Kron reduction removes, and the excluded neighbourhood is the smallest that certifies within budget rather than the smallest that exists. Branch-and-bound is budget-limited, so `unknown` is possible and is never reported as a refutation. Measured compute is 28 s for the sample-based reproduction, 215 to 533 s for 2-D certified training, 836 s for structural retraining, and $1{,}344$ to $8{,}875$ s for the 4-D runs on one CPU, which makes the dimensional wall practical as well as theoretical.

Broader impact. A satisfaction fraction over samples cannot support a decision about raising a renewable cap, since it says nothing about the states between the samples, while a region certificate names the set of post-fault states the loop returns to synchronism. The next step is compositional, certifying interacting subregions and combining their certificates.

References

H.-D. Chiang. Study of the existence of energy functions for power systems with losses. *IEEE Transactions on Circuits and Systems*, 36(11):1423–1429, 1989.

W. Cui and B. Zhang. Lyapunov-regularized reinforcement learning for power system transient stability. *IEEE Control Systems Letters*, 6:974–979, 2022.

H. Zhang, T.-W. Weng, P.-Y. Chen, C.-J. Hsieh, and L. Daniel. Efficient neural network robustness certification with general activation functions. In *Advances in Neural Information Processing Systems*, 2018.

K. Xu, Z. Shi, H. Zhang, Y. Wang, K.-W. Chang, M. Huang, B. Kailkhura, X. Lin, and C.-J. Hsieh. Automatic perturbation analysis for scalable certified robustness and beyond. In *Advances in Neural Information Processing Systems*, 2020.

Z. Shi, Q. Jin, J. Z. Kolter, S. Jana, C.-J. Hsieh, and H. Zhang. Neural network verification with branch-and-bound for general nonlinearities. *arXiv preprint arXiv:2405.21063*, 2024.

Z. Shi, H. Li, C.-J. Hsieh, and H. Zhang. Certified training with branch-and-bound: a case study on Lyapunov-stable neural control. *arXiv preprint arXiv:2411.18235*, 2024.

T. Athay, R. Podmore, and S. Virmani. A practical method for the direct analysis of transient stability. *IEEE Transactions on Power Apparatus and Systems*, PAS-98:573–584, 1979.

Y.-C. Chang, N. Roohi, and S. Gao. Neural Lyapunov control. In *Advances in Neural Information Processing Systems*, 2019.

Availability of code and data

All numbers are regenerated from saved per-seed records and trained checkpoints by a single script that also asserts the headline values it plots, including the 0.4991 envelope fraction and the $+35.98$ maximum Lie derivative, so a stale figure fails loudly rather than shipping. Code, records, and checkpoints are available from the author on request, together with environment pins for the CPU-only setup and the one documented numerical workaround. No data beyond the public 39-bus model is used.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to re-run every numerical claim from the saved per-seed records through the asserting script described above so the numbers are checked by execution rather than by reading, and to verify citations against the published record. The authors directed the work and take responsibility for all content.